

Data-Efficient and Low-Complexity Acoustic Scene Classification



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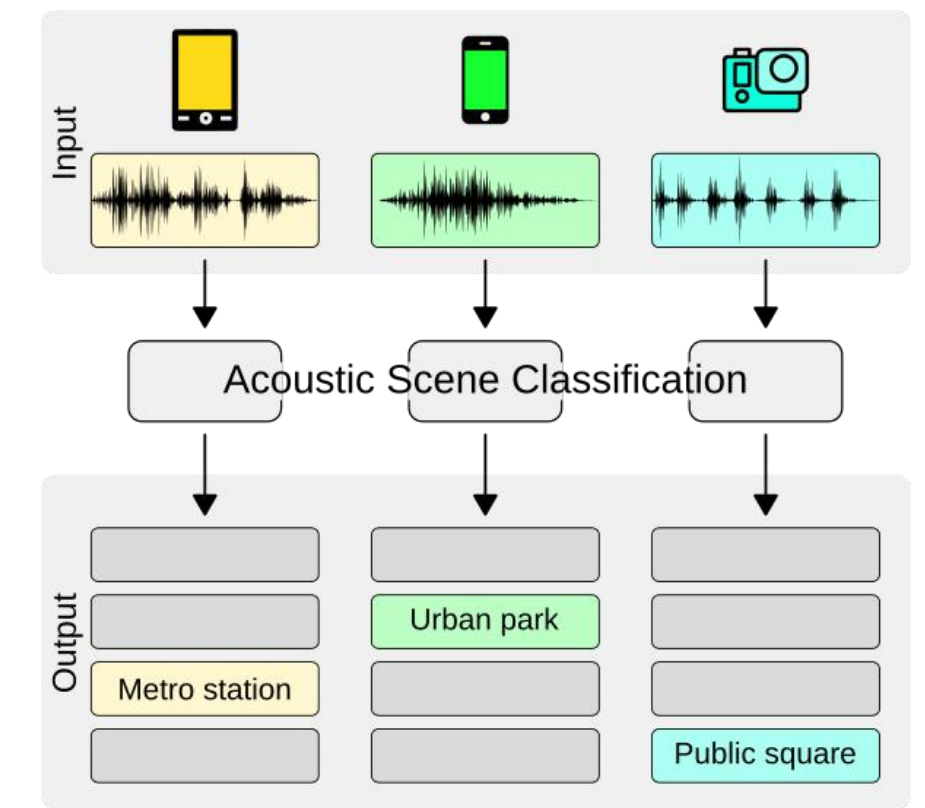
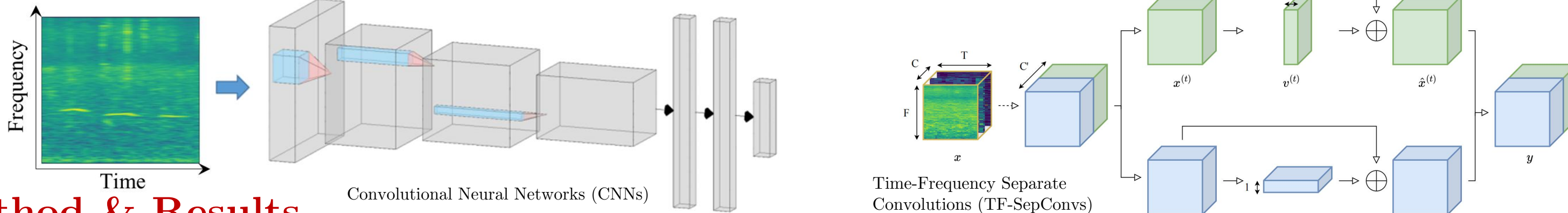


Abstract

Acoustic Scene Classification (ASC) is a fundamental task in machine listening, enabling devices to recognize environmental contexts from audio signals. However, real-world deployment faces critical challenges: domain shift in devices, low-complexity requirement and data efficiency. Our project focuses on addressing these issues by creating low-complexity ASC models trained on small subsets of the TAU Urban Acoustic Scenes dataset while maintaining high accuracy.

Introduction

Acoustic Scene Classification (ASC) is a task to recognize the environment in which an audio recording was captured, such as streets, parks or airports. Traditional approaches to ASC typically base on **Convolutional Neural Networks (CNNs)**. However, promoting in both computational load and output efficiency presents considerable challenges, often involving extensive system structure innovation and accuracy optimization efforts. The **Time-Frequency Separate Network (TF-SepNet)** is a CNN architecture consist of **Time-Frequency Separate Convolutions (TF-SepConvs)**, for separating the processing of feature maps in CNNs specifically optimized for low-complexity ASC tasks, which encourages the development of efficient models capable of maintaining high performance despite reduced training data, advancing the practical applicability and scalability of ASC systems in real world.



Method & Results

Knowledge Distillation (KD)

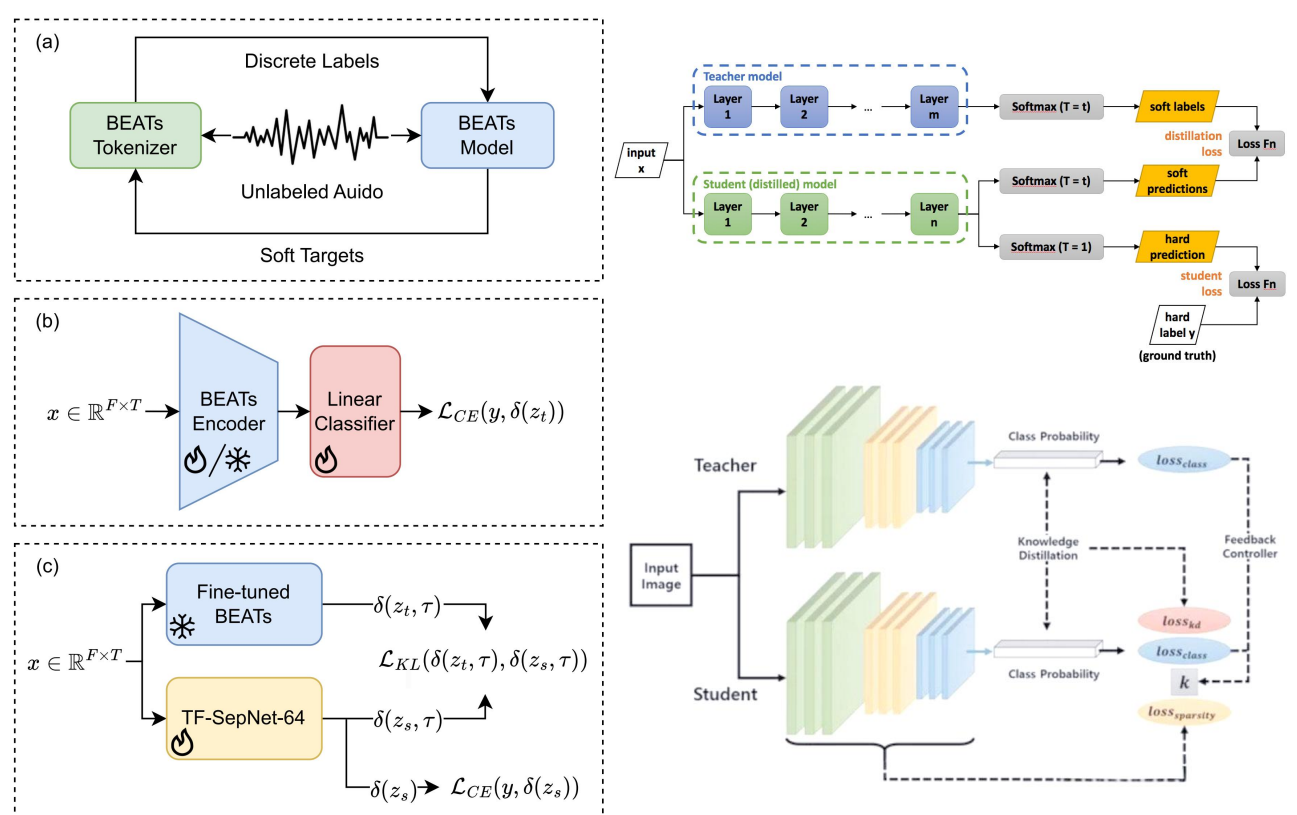


Figure 1. Knowledge distillation with self-supervised teachers. (a) Self-supervised pre-training BEATs on AudioSet. (b) Fine tuning pre-trained BEATs on ASC dataset. (c) Distilling knowledge from fine-tuned BEATs to TF-SepNet-64. Snowflake icon indicates that the parameters of the corresponding part are frozen, while Flame icon indicates the opposite.

Pruning

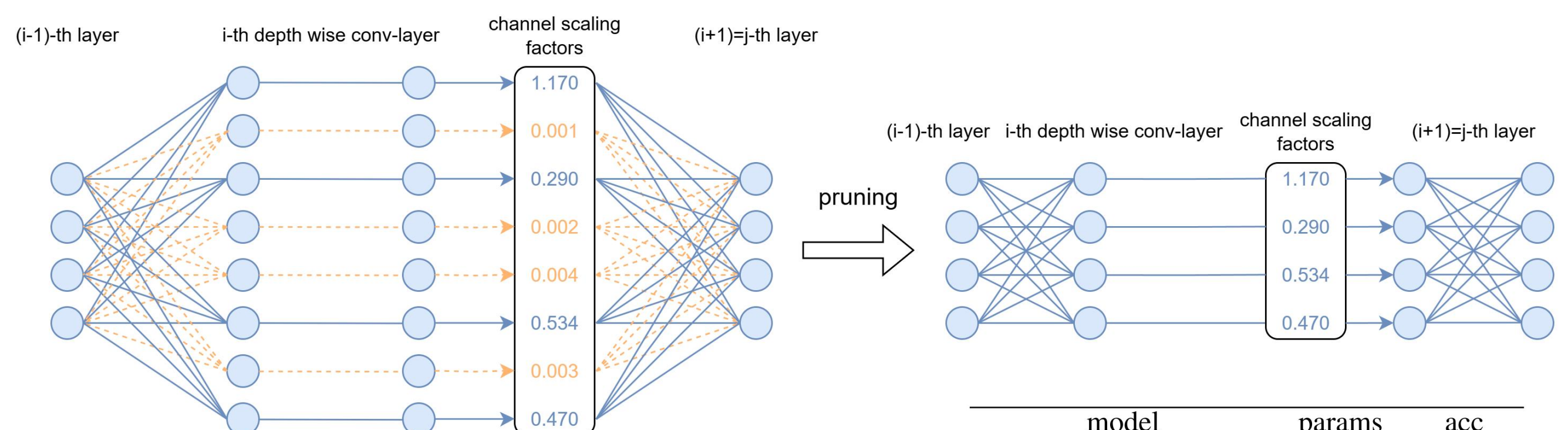


Figure 2. We associate a scaling factor (reused from a batch normalization layer) with the channel in depth convolutional layers in each block. Sparsity regularization is imposed on these scaling factors during training to automatically identify unimportant channels. The channels with small scaling factor values (in orange color) will be pruned (left side). After pruning, we obtain compact models (right side), which are then fine-scaling to achieve comparable(or even higher) accuracy as normally trained full network. The results are shown in the table.

model	params	acc
TFSepNet	123k	44.5%
cp_mobile	126k	42.1%
cp_mobile(prune)	47k	43.3%
sep_cp_mobile	123k	45.1%
sep_cp_mobile(prune)	55k	46.0%

Domain Adversarial Training

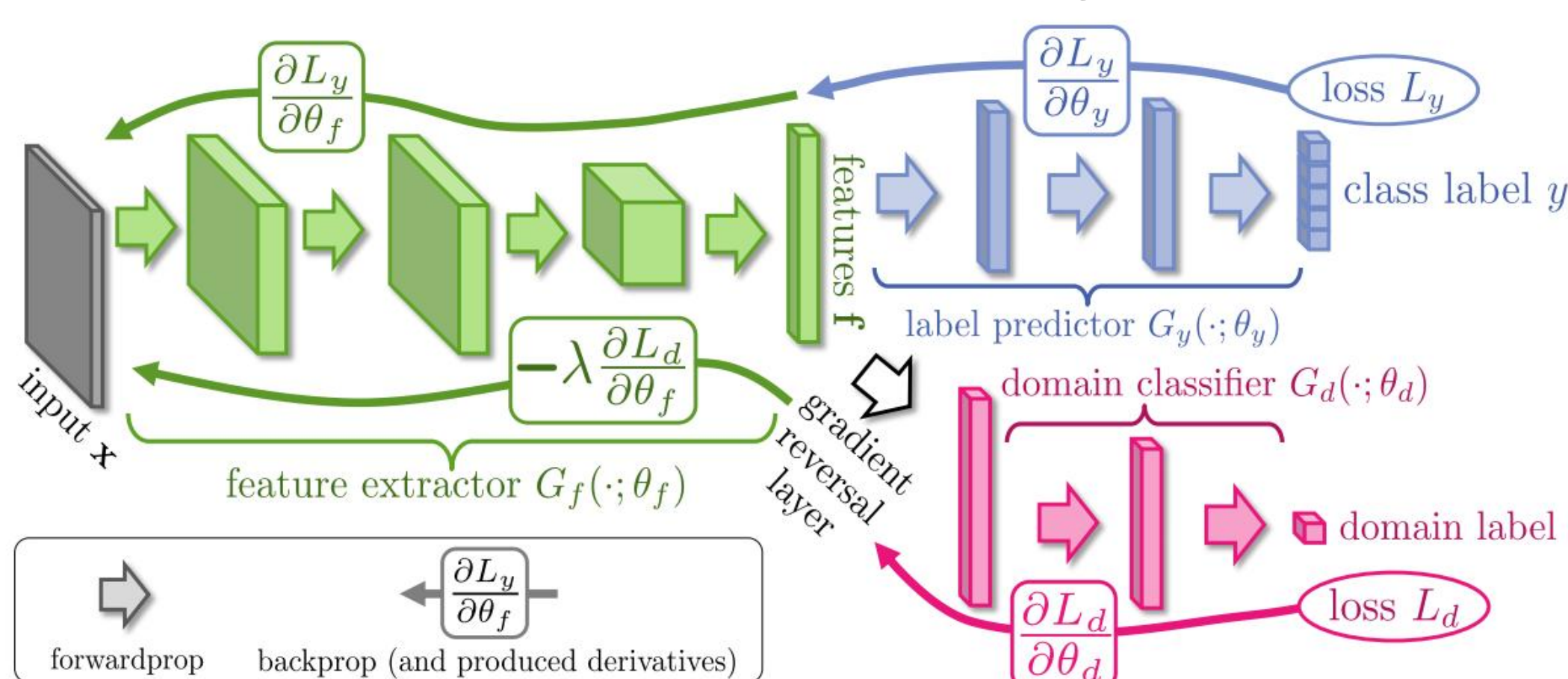


Figure 3. The proposed architecture includes a deep feature extractor (green) and a deep label predictor (blue), which together form a standard feed-forward architecture. Unsupervised domain adaptation is achieved by adding a domain classifier (red) connected to the feature extractor via a gradient reversal layer that multiplies the gradient by a certain negative constant during the backpropagation based training. Gradient reversal ensures that the feature distributions over the two domains are made similar (as indistinguishable as possible for the domain classifier), thus resulting in the domain-invariant features.

Lightweight Model Design

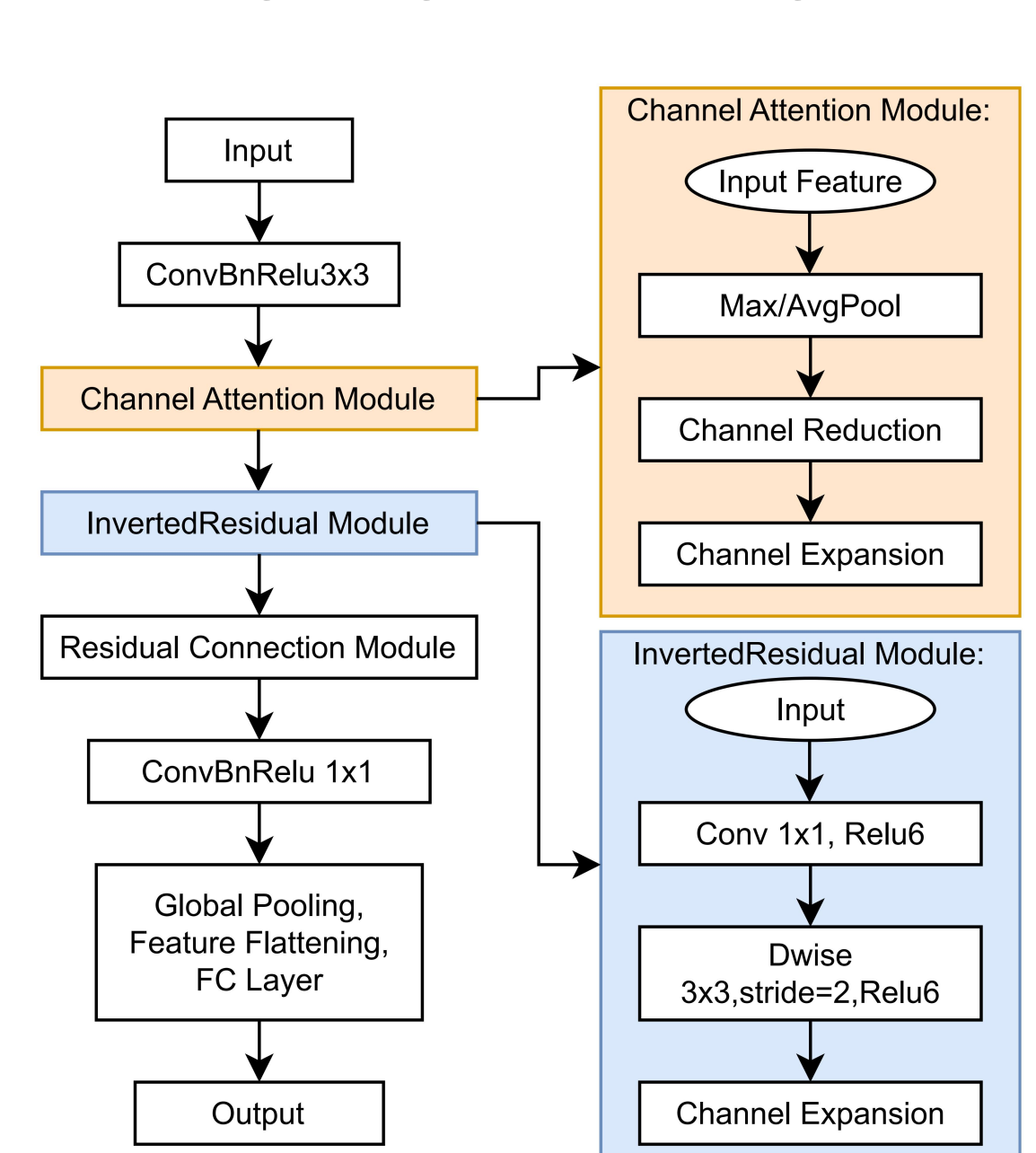


Figure 6. This architecture employs an inverted residual structure with depthwise separable convolutions to construct a lightweight and efficient model. In order to maximum the output of the inverted residual block, a bottleneck-shaped attention module is applied before it, which optimizes the allocation of feature channel resources, thereby providing higher quality and more focused input features for the following depthwise convolution.

Performance in seen&unseen devices

Model	Unseen	Seen	Target/Source Ratio (%)
TF-SepNet	42.13	47.05	89.50
TF-SepGrlNet	45.50	47.87	95.05
TF-SepDualNet*	61.63	63.35	97.20

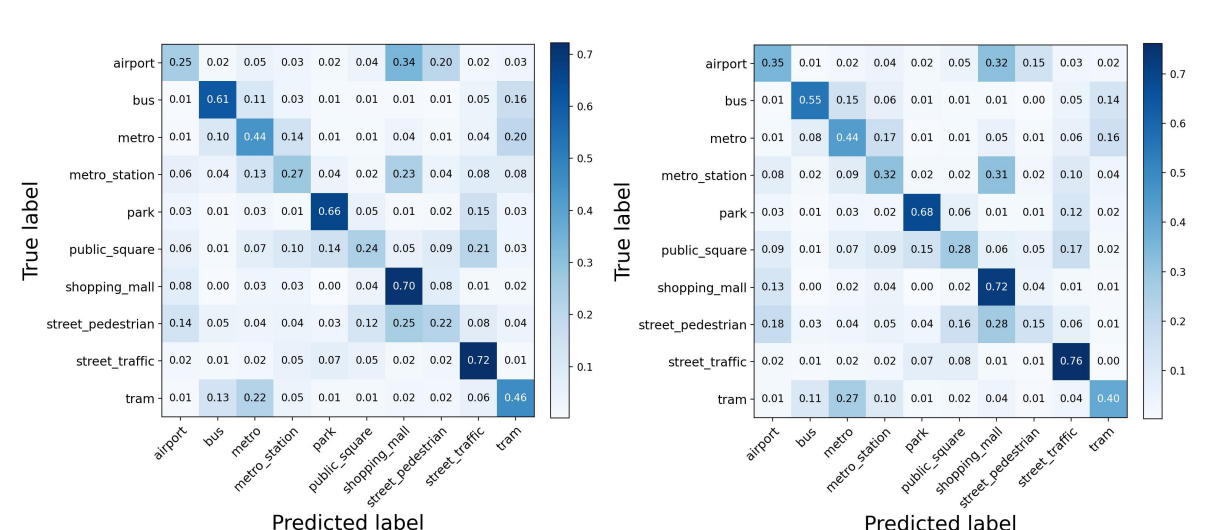


Figure 4. Domain Shift results and label outputs on the test set of TAU Urban Acoustic Scene 2022 Mobile development dataset. Target/Source Ratio denotes the average accuracy of which on unseen domains should achieve an accuracy of seen domain.

Cross City&Device Accuracy during Training

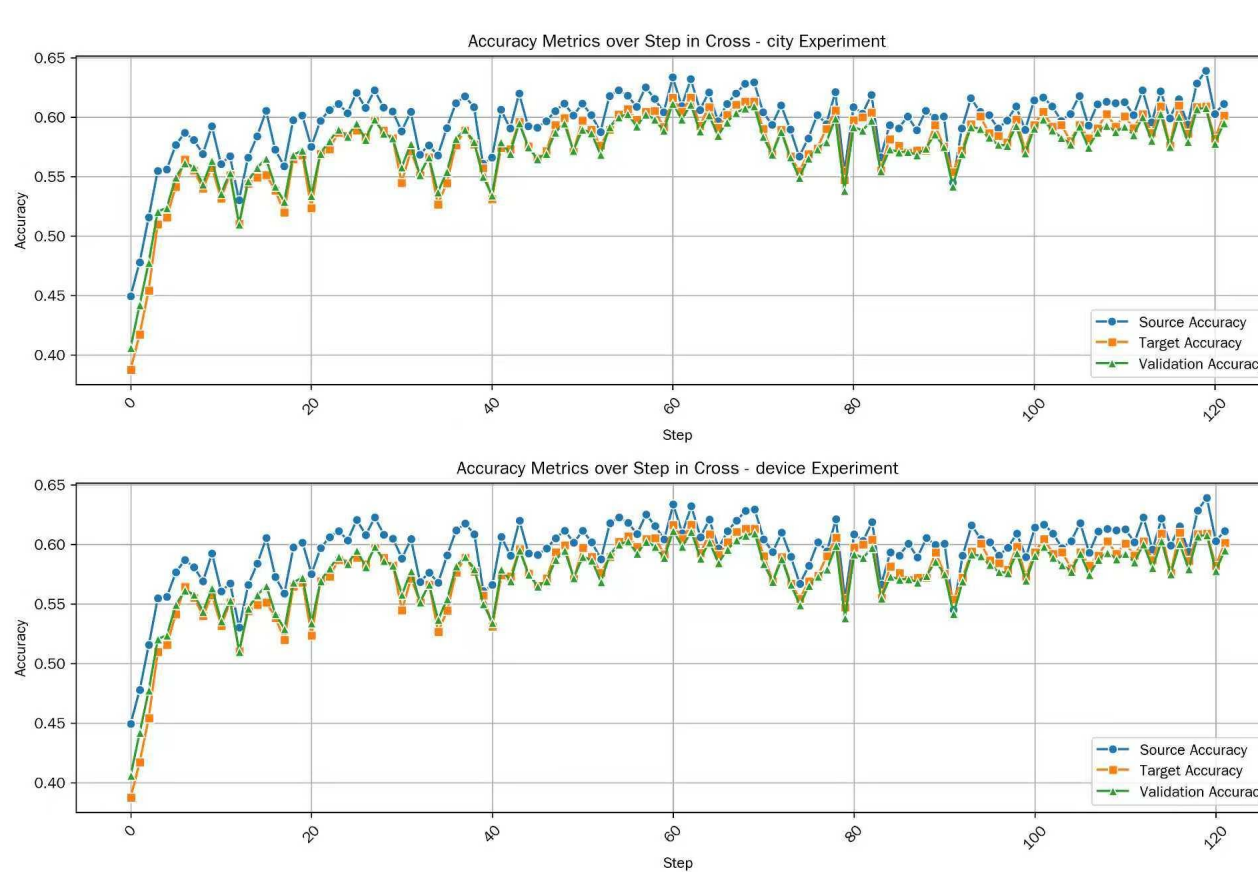


Figure 5. Accuracy over steps in city&device experiment. The test accuracy refers to the final model performance on an independent test set. The results in city labels are shown in the table with “* ”.

Method	Params (k)	MMACs	Accuracy (%)
DCASE Baseline	61.15	29.42	53.24
student_KD	29.93	13.12	54.47

Figure 7. As a result, student_KD could achieve an accuracy of 54.45%, with only 48.94% parameters (29.93k) and 44.60% flops (13.12MMACs) using knowledge distillation.

Conclusions

In this project, we propose a comprehensive framework for data-efficient and low-complexity Acoustic Scene Classification (ASC) by integrating: Knowledge Distillation, Structured Pruning, Domain Adversarial Training and Lightweight Model Design. Our ASC model achieves 54.45% accuracy on the TAU 2022 Mobile dataset with only 29.93K parameters and 13.12 MMACs and sustained improvement in accuracy during the pruning operation, demonstrating a favorable trade-off between performance and complexity. Additionally, we still acheive a high accuracy over steps in both device and city experiments, validating the effectiveness of our approach in enabling practical ASC deployment on resource-constrained devices in the future.